

Logical Volume Management(LVM)

Q.1 Create two 200MB partitions using fdisk or parted, then use the lsblk command to verify their creation.

Ans- Step 1-Check disk: lsblk

Step 2- Create partitions: fdisk /dev/sdb

Step 3-Verify- lsblk

```
root@myfoodiebox:~  
root@myfoodiebox:~# lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda          8:0    0   60G  0 disk  
├─sda1       8:1    0    1M  0 part  
├─sda2       8:2    0    2G   0 part /boot  
└─sda3       8:3    0   58G  0 part  
   └─cs-root 253:0    0 36.4G  0 lvm  /  
   └─cs-swap 253:1    0  3.9G  0 lvm  [SWAP]  
   └─cs-home 253:2    0 17.7G  0 lvm  /home  
sdb          8:16    0    2G   0 disk  
sr0         11:0    1  9.7G  0 rom   /run/media/root/CentOS-Stream-10-BaseOS-x86_64  
root@myfoodiebox:~# fdisk /dev/sdb  
bash: fdisk/dev/sdb: No such file or directory  
root@myfoodiebox:~# fdisk /dev/sdb  
  
Welcome to fdisk (util-linux 2.40.2).  
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.  
  
Device does not contain a recognized partition table.  
Created a new DOS (MBR) disklabel with disk identifier 0x407dd647.  
  
Command (m for help): p  
Disk /dev/sdb: 2 GiB, 2147483648 bytes, 4194304 sectors  
Disk model: VMware Virtual S  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: dos  
Disk identifier: 0x407dd647  
  
Command (m for help): p  
Disk /dev/sdb: 2 GiB, 2147483648 bytes, 4194304 sectors  
Disk model: VMware Virtual S  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 512 bytes  
I/O size (minimum/optimal): 512 bytes / 512 bytes  
Disklabel type: dos  
Disk identifier: 0x407dd647  
  
Command (m for help): w  
The partition table has been altered.  
Calling ioctl() to re-read partition table.  
Syncing disks.  
  
root@myfoodiebox:~# lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
sda          8:0    0   60G  0 disk  
├─sda1       8:1    0    1M  0 part  
├─sda2       8:2    0    2G   0 part /boot  
└─sda3       8:3    0   58G  0 part  
   └─cs-root 253:0    0 36.4G  0 lvm  /  
   └─cs-swap 253:1    0  3.9G  0 lvm  [SWAP]  
   └─cs-home 253:2    0 17.7G  0 lvm  /home  
sdb          8:16    0    2G   0 disk  
sr0         11:0    1  9.7G  0 rom   /run/media/root/CentOS-Stream-10-BaseOS-x86_64  
root@myfoodiebox:~#
```

Q.2 Initialize both partitions as physical volumes using the pvcreate command and confirm their status with pvs.

Ans- Step 1- pvcreate /dev/sdb1 /dev/sdb2

Step 2-Verify: pvs

```
root@myfoodiebox:~# pvcreate /dev/sdb1
Physical volume "/dev/sdb1" successfully created.
root@myfoodiebox:~# pvcreate /dev/sdb2
Physical volume "/dev/sdb2" successfully created.
root@myfoodiebox:~# pvs
PV          VG Fmt Attr PSize  PFree
/dev/sda3   cs  lvm2 a--  <58.00g    0
/dev/sdb1    lvm2 ---  200.00m 200.00m
/dev/sdb2    lvm2 ---  200.00m 200.00m
```

Q.3 Create a new volume group named musicvg using the two physical volumes, then display its details with vgdisplay.

Ans- Step 1- vgcreate musicvg /dev/sdb1 /dev/sdb2

Step 2- Verify: vgdisplay musicvg

```
root@myfoodiebox:~# vgcreate musicvg /dev/sdb1
Volume group "musicvg" successfully created
root@myfoodiebox:~# vgdisplay musicvg
--- Volume group ---
VG Name                musicvg
System ID
Format                  lvm2
Metadata Areas          1
Metadata Sequence No    1
VG Access               read/write
VG Status               resizable
MAX LV                  0
Cur LV                 0
Open LV                 0
Max PV                  0
Cur PV                 1
Act PV                  1
VG Size                 196.00 MiB
PE Size                 4.00 MiB
Total PE                 49
Alloc PE / Size         0 / 0
Free PE / Size          49 / 196.00 MiB
VG UUID                 FK6ZtP-Jao7-jCfm-g8m2-OEPF-dhjY-8hJz1R

root@myfoodiebox:~#
```

Q.4 Create a logical volume called playlistlv of size 300MB inside musicvg, format it with ext4, and mount it to /mnt/playlist.

Ans- Step 1- Create LV: `lvcreate -L 300M -n playlistlv musicvg`

Step 2- Format: `mkfs.ext4 /dev/musicvg/playlistlv`

Step 3- Create Mount point: `mkdir -p /mnt/playlist`

Step 4- Mount: `mount /dev/musicvg/playlistlv/mnt/playlist`

Step 5- Verify: `df -h`

```
[root@localhost ~]# lvcreate -L 300M -n playlistlv musicvg
Logical volume "playlistlv" created.
[root@localhost ~]# mkfs.ext4 /dev/musicvg/playlistlv
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 307200 1k blocks and 76912 inodes
Filesystem UUID: fd582e6e-f8c6-446a-810a-fbaf7c460eca
Superblock backups stored on blocks:
    8193, 24577, 40961, 57345, 73729, 204801, 221185

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# mkdir -p /mnt/playlist
[root@localhost ~]# mount /dev/musicvg/playlistlv/mnt/playlist
mount: /dev/musicvg/playlistlv/mnt/playlist: can't find in /etc/fstab.
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0   4.0M   0% /dev
tmpfs           871M   0   871M   0% /dev/shm
tmpfs           349M  7.2M  342M   3% /run
/dev/mapper/cs-root 17G  5.9G  12G  35% /
/dev/sda1       960M  349M  612M  37% /boot
tmpfs           175M  96K  175M   1% /run/user/1000
/dev/sr0        11G   11G    0 100% /run/media/user/CentOS-Stream-9-BaseOS-x86_64
```

**Q.5 Resize playlistlv by adding 50MB more space using lvextend, then expand the filesystem and show the updated size with df -h.

Hint: Use resize2fs after lvextend to update the filesystem size.**

Ans- Step 1-Extend Lv: `lvextend -L +50M /dev/musicvg/playlistlv`

Step 2-Resize Filesystem: `resize2fs /dev/musicvg/playlistlv`

Step 3- verify new size: `df -h`

```
[root@localhost ~]# lvextend -L +50M /dev/musicvg/playlistlv
Rounding size to boundary between physical extents: 52.00 MiB.
Size of logical volume musicvg/playlistlv changed from 300.00 MiB (75 extents)
to 352.00 MiB (88 extents).
Logical volume musicvg/playlistlv successfully resized.
[root@localhost ~]# resize2fs /dev/musicvg/playlistlv
resize2fs 1.46.5 (30-Dec-2021)
Resizing the filesystem on /dev/musicvg/playlistlv to 360448 (1k) blocks.
The filesystem on /dev/musicvg/playlistlv is now 360448 (1k) blocks long.

[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        4.0M   0   4.0M   0% /dev
tmpfs           871M   0   871M   0% /dev/shm
tmpfs           349M  7.2M  342M   3% /run
/dev/mapper/cs-root 17G  5.9G  12G  35% /
/dev/sda1       960M  349M  612M  37% /boot
tmpfs           175M  96K  175M   1% /run/user/1000
/dev/sr0        11G   11G    0 100% /run/media/user/CentOS-Stream-9-BaseOS-x86_64
```